

PRIMARY AI FINDING	Model confidence	99.1% (calibrated)
Glioma	Risk level	High
Confident - single-class prediction set	Prediction set	Glioma
	Coverage guarantee	99%

IMPORTANT CAVEATS

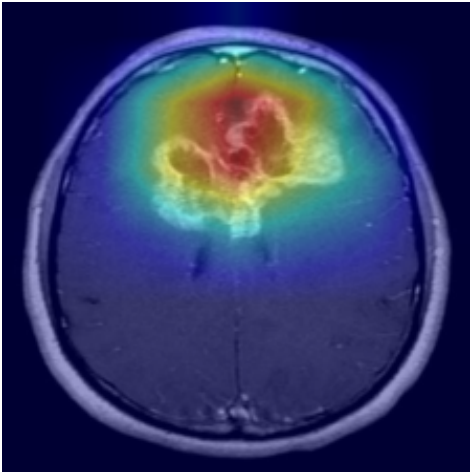
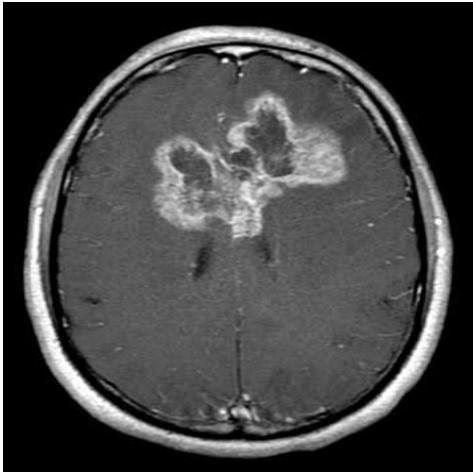
- js_divergence = 0.2606 exceeds the fitted threshold 0.0264; this image does not resemble the brain MRI distribution the model was trained on.

PROBABILITY DISTRIBUTION

Class	Probability		In set
Glioma	99.10%		yes
Meningioma	0.40%		-
Pituitary	0.37%		-
No Tumor	0.13%		-

Probabilities are the stacking meta-learner's posterior. Values are temperature-calibrated. The 'In set' column marks classes retained by the conformal prediction set at the stated coverage level.

EXPLAINABILITY



Original scan (resized)

Grad-CAM attention overlay

The heatmap shows which image regions most influenced the model's output. It is an interpretability aid and must not be read as tumour segmentation, a boundary, or a measurement.

AI SUMMARY

The model classified this scan as 'Glioma' with 99.1% confidence.

Generated from deterministic templates. No AI-authored narrative has been approved for this finding and uncertainty band.

CONDITION OVERVIEW

Glioma is a broad term for tumours arising from glial cells ? the supporting cells of the central nervous system ? rather than from neurons themselves. The group spans a wide range of biological

behaviour, from slow-growing lesions to aggressive high-grade tumours. Because behaviour varies so much within the group, "glioma" alone is not a complete diagnosis: type and grade determine prognosis and management.

Current typing and grading follow the WHO Classification of Tumours of the Central Nervous System, 5th edition, which integrates histological appearance with molecular markers. Grading terminology changed from the 4th edition, so older literature is not directly comparable.

LIMITATIONS FOR THIS FINDING

An image-only classifier assigns a broad tumour category. It does not establish grade, molecular subtype, margins, or volume, and none of those can be inferred reliably from a single 2D slice. In this project's own evaluation, glioma was the weakest class for recall – the single-model baseline recovered 0.81 of true gliomas, with the missed cases tending to be absorbed into the meningioma class. Both tumour types can enhance similarly on T1, so that confusion is radiologically plausible rather than random. The stacking ensemble improved glioma recall to 0.90, but the class remains the least reliable of the four.

Consequently a glioma output from this system should be read as "features that the model associates with glioma are present", not as a confirmed tumour type.

POTENTIAL FURTHER EVALUATION

Investigations that a clinician may consider discussing, depending on presentation:

- Clinical and neurological examination to correlate imaging with symptoms and deficits
- Dedicated contrast-enhanced MRI using a brain-tumour protocol, which provides information a single unenhanced slice cannot
- Specialist review by neurology, neurosurgery or neuro-oncology
- Histopathological and molecular characterisation where tissue is obtained, since typing and grading under the current WHO classification depend on molecular markers
- Additional imaging or functional studies where surgical planning is being considered

CLINICAL CONSIDERATIONS

- Symptoms depend heavily on tumour location, size and growth rate rather than on tumour type alone.
- An imaging appearance suggestive of glioma does not establish grade, and grade is the dominant driver of prognosis and treatment intensity.
- Management is multidisciplinary; decisions are not made from imaging in isolation.
- Incidental findings occur. Correlation with the clinical picture is essential.

FOLLOW-UP CONSIDERATIONS

- Specialist review to determine whether and when further imaging is appropriate
- Comparison against any prior imaging, where available, since change over time carries more information than a single study
- Reassessment if new or progressive neurological symptoms develop

WARNING SIGNS - SEEK PROMPT MEDICAL ATTENTION

Symptoms that generally warrant prompt medical assessment rather than waiting for a routine appointment:

- Sudden severe headache, particularly if different in character from previous headaches
- New seizure activity
- Loss of consciousness
- Sudden weakness, numbness or difficulty speaking
- Significant or rapidly progressing visual change
- Confusion or a marked change in mental state
- Persistent vomiting with headache

These are general neurological red flags, not glioma-specific, and apply regardless of what any AI system reports.

TREATMENT INFORMATION (FOR CLINICIAN REVIEW)

For clinician review. Not a prescription, and no medicines or doses are specified.

Management depends on factors including tumour type and grade, location, size, growth characteristics, the patient's symptoms, age and overall clinical condition. Broad categories described in the literature include:

- Observation with interval imaging in selected situations
- Surgical management, including biopsy for diagnosis and resection where feasible
- Radiation-based approaches
- Systemic therapy
- Supportive and symptom-directed care

Which categories apply, in what sequence, and with what intent is a specialist decision made after diagnostic confirmation and full clinical evaluation.

This is general reference information, not a prescription. No medicine, dose or schedule is recommended by this system.

EVIDENCE AND REFERENCES

#	Source	Publisher / reference
1	Central Nervous System Tumors Treatment (PDQ) - Health Professional Version	National Cancer Institute (NCI) - https://www.cancer.gov/types/brain/hp/adult-brain-treatment-pdq
2	Adult Central Nervous System Tumors Treatment (PDQ) - Patient Version	National Cancer Institute (NCI), U.S. National Institutes of Health - https://www.cancer.gov/types/brain/patient/adult-brain-treatment-pdq
3	Brain and Spinal Cord Tumors	National Institute of Neurological Disorders and Stroke (NINDS), NIH - https://www.ninds.nih.gov/health-information/disorders/brain-and-spinal-cord-tumors
4	WHO Classification of Tumours of the Central Nervous System, 5th edition	World Health Organization / IARC, 2021 - https://www.iarc.who.int/

The clinical content in this report is pending verification against the primary sources by a qualified reviewer.

AI LIMITATIONS AND INTENDED USE

This AI-generated report is for clinical decision support only. It must not replace professional medical diagnosis. All findings should be validated by a qualified radiologist or specialist physician.